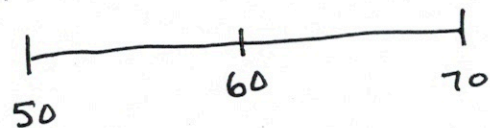
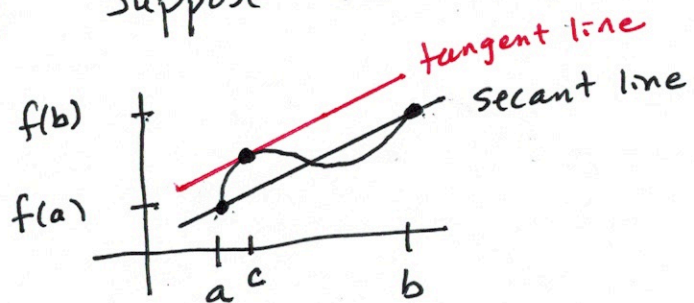


Section 4.2 The Mean Value Theorem

drive 50 mph $\longrightarrow$ 70 mph



MVT Suppose $f(x)$ is continuous on $[a, b]$



there is a number c in (a, b) so that

$$\underbrace{f'(c)}_{\text{slope of tangent line}} = \underbrace{\frac{f(b) - f(a)}{b - a}}_{\text{slope of secant line}}$$

Ex ① Find the value c guaranteed by MVT

$$f(x) = x^3 - 2x^2 + x + 1 \text{ on } [0, 2]$$

$$\begin{aligned} \text{slope of secant line} &= \frac{f(2) - f(0)}{2 - 0} \\ &= \frac{2^3 - 2(2)^2 + 2 + 1 - 1}{2} \\ &= \frac{2}{2} \\ &= 1 \checkmark \end{aligned}$$

$$f'(x) = 3x^2 - 2 \cdot 2x + 1$$

$$1 = 3c^2 - 4c + 1$$

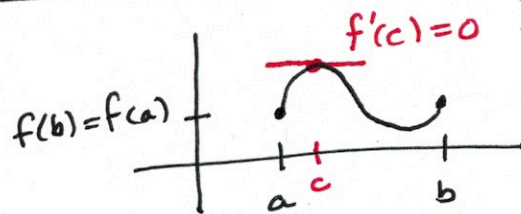
$$0 = 3c^2 - 4c$$

$$0 = c(3c - 4)$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ c \neq 0 & & 3c - 4 = 0 \end{array}$$

$$\boxed{c = \frac{4}{3}}$$

Rolle's Theorem: MVT with $f(b) = f(a)$



$$\frac{f(b) - f(a)}{b - a} = \frac{0}{b - a} = 0$$

Ex ② Verify Rolle's Theorem

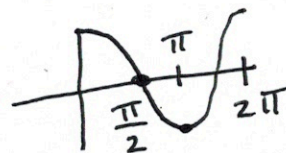
$$f(x) = \sin\left(\frac{x}{2}\right) \text{ on } [0, 2\pi]$$

$$\frac{f(2\pi) - f(0)}{2\pi - 0} = \frac{\sin(\pi) - \sin(0)}{2\pi} = 0 \checkmark$$

$$f'(x) = \cos\left(\frac{x}{2}\right) \cdot \left(\frac{1}{2}\right)$$

$$0 = \cos\left(\frac{c}{2}\right) \cdot \frac{1}{2}$$

$$\downarrow$$
$$0 = \cos\left(\frac{c}{2}\right)$$



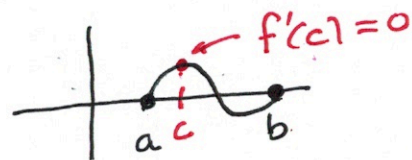
$$\frac{c}{2} = \frac{\pi}{2} \rightarrow \boxed{c = \pi}$$

EX ③ Prove that $\sin x = 2x - 1$ has only one root.

$$\sin x - 2x + 1 = 0$$

$$f(x) = \sin x - 2x + 1$$

Suppose $f(x)$ has 2 roots: a and b .

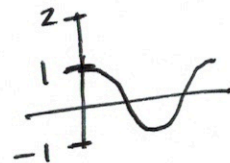


$$f'(x) = \cos x - 2$$

$$0 = \cos(c) - 2$$

$$2 = \cos(c)$$

Impossible!



So, it cannot have multiple roots!

Ex (4) Two runners start a marathon at the same time & finish at the same time.

Prove that there was a moment when they ran the same pace.

$f(t)$ = distance for Frank

$h(t)$ = distance for Helen

start $f(a) = 0$ and $h(a) = 0$

end $f(b) = 26.2$ and $h(b) = 26.2$

Let $d(t)$ be the distance between them.

$$d(t) = f(t) - h(t)$$

$$d'(t) = f'(t) - h'(t)$$

$$0 = f'(c) - h'(c)$$

$$h'(c) = f'(c)$$

Same rate!

